HR SQL Analysis

MySQL | HR Data Analysis

Project Overview
An analysis of HR data using SQL to answer business questions related to workforce, compensation, performance, hiring, demographics, and attrition.

Executive Summary

Objective

Analyzed employee data using SQL to derive actionable insights for HR and management decision-making.

Project Scope	HR Analytics
Business Questions	22
Tool	MySQL
Analysis Areas	Workforce, Compensation, Performance, Hiring, Demographics, Attrition

Analysis Focus

• Workforce size and employee composition

• Salary and compensation patterns

• Employee performance

• Hiring trends

• Employee demographics and locations

• Attrition

SQL Techniques Used

SELECT • WHERE • GROUP BY • ORDER BY • JOINs • CASE WHEN • Aggregate Functions • Date Functions • CTEs • Window Functions

Business Questions & Uses

Q1. What is the total number of employees currently registered in the organization?

```
SELECT COUNT(*) as total_employees
FROM employees;
```

total_employees
10000

Business Use: Provides the total employee count as a baseline for workforce planning and HR analysis.

Q2. How many employees currently have an "Active" employment status?

```
SELECT COUNT(*)
FROM employees
WHERE employment_status = 'Active';
```

COUNT(*)
7018

Business Use: Helps HR determine the current workforce available for staffing decisions.

Q3. What percentage of employees have left the company (resigned or terminated)?

```
SELECT ROUND(SUM(CASE
                                WHEN employment_status IN ('Resigned', 'Terminated')
        THEN 1
        ELSE 0
        END
)/ COUNT(*) * 100.0,2) as attrition_rate
FROM employees;
```

attrition_rate
29.82

Business Use: Helps management identify whether **employee attrition is becoming a retention concern**.

Q4. What is the average salary across all employees?

```
SELECT ROUND(AVG(salary)) as avg_salary
FROM employees;
```

avg_salary
136302

Business Use: Provides a baseline for salary budgeting and overall compensation planning.

Q5. What is the average salary in each department, and which department pays the most?

```
SELECT d.department_name, ROUND(AVG(e.salary)) as avg_salary
FROM employees e
JOIN departments d
```

Business Questions & Uses

ON d.department_id = e.department_id

GROUP BY d.department_name

ORDER BY avg_salary DESC;

department_name	avg_salary
Operations	138354
Quality Assurance	138233
Procurement	138084
Administration	137563
Legal	137261
Customer Support	137035
Finance	136759
Marketing	136536
Information Technology	134483
Research and Development	133944
Sales	133910
Human Resources	133440

Business Use: Helps management identify departments with higher compensation requirements for budget allocation.

Q6. What is the average salary across different job designations/roles?

SELECT designation, ROUND(AVG(salary)) as avg_salary

FROM employees

GROUP BY designation;

designation	avg_salary
SEO Specialist	133756
Software Engineer	131130
Accountant	135623
Auditor	136901
Operations Manager	138693
R&D Engineer	131590
Senior Software Engineer	127775
Office Manager	143042
Legal Advisor	136409
Legal Counsel	138330
Marketing Executive	134525

Business Use: Helps HR identify high-cost job roles for salary benchmarking and compensation planning.

Q7. Average Performance Rating

SELECT ROUND(AVG(performance_rating),2) as avg_performance_rating

FROM performance;

avg_performance_rating
2.99

Business Use: Helps management assess the organization's overall employee performance level.

Q8. Who are the employees with the highest performance ratings (top performers)?

SELECT e.employee_id, e.employee_name, p.performance_rating

FROM employees e

JOIN performance p

Business Questions & Uses

```
ON e.employee_id = p.employee_id
ORDER BY p.performance_rating DESC
LIMIT 10;
```

employee_id	employee_name	performance_rating
4438	Saanvi Singh	5.00
3985	Kavya Agarwal	5.00
7709	Sanjay Bose	5.00
4693	Kavya Joshi	5.00
1727	Vivaan Bose	5.00
3799	Priya Bose	5.00
6748	Rohan Agarwal	5.00
3949	Ananya Joshi	5.00
3401	Sanjay Chopra	5.00
5402	Swati Mehta	5.00

Business Use: Helps HR identify top performers for recognition, rewards, and career development.

Q9. How is bonus amount distributed across employees?

```
SELECT CASE
                WHEN bonus <=10000 THEN '0-10000'
    WHEN bonus <=20000 THEN '10001-20000'
    WHEN bonus <=30000 THEN '20001-30000'
    WHEN bonus <=40000 THEN '30001-40000'
     ELSE'40001-50000'
  END as bonus_range,
COUNT(*) as employee_count
FROM performance
GROUP BY bonus_range
ORDER BY bonus_range;
```

bonus_range	employee_count
0-10000	1953
10001-20000	1997
20001-30000	2014
30001-40000	2022
40001-50000	2014

Business Use: Helps management evaluate how bonus payments are distributed for incentive planning.

Q10. How many employees work in each department?

```
SELECT d.department_name, COUNT(e.employee_id) as number_of_employees
FROM departments d
LEFT JOIN employees e
ON d.department_id = e.department_id
GROUP BY d.department_name
ORDER BY number_of_employees DESC;
```

Business Questions & Uses

department_name	number_of_employees
Finance	886
Information Technology	858
Procurement	845
Marketing	841
Administration	833
Human Resources	828
Quality Assurance	827
Research and Development	821
Operations	818
Customer Support	817
Sales	814
Legal	812

Business Use: Helps management identify staffing levels across departments for resource allocation.

Q11. What is the distribution of male to female employees in the company?

```
SELECT gender, COUNT(*) as total_employees, ROUND((COUNT(*)/(SELECT COUNT(*) FROM employees))*100,2) as percentage
FROM employees
GROUP BY gender;
```

gender	total_employees	percentage
Male	4985	49.85
Female	5015	50.15

Business Use: Helps HR monitor gender diversity and support workforce diversity initiatives.

Q12. Which age group has the highest concentration of employees?

```
SELECT CASE
            WHEN age BETWEEN 20 AND 29 THEN '20-29'
    WHEN age BETWEEN 30 AND 39 THEN '30-39'
    WHEN age BETWEEN 40 AND 49 THEN '40-49'
    ELSE '50-59'
    END as age_group,
    COUNT(*) as num_of_employees
FROM employees
GROUP BY age_group
ORDER BY age_group DESC;
```

age_group	num_of_employees
50-59	2533
40-49	2547
30-39	2578
20-29	2342

Business Use: Helps HR understand the age composition of employees for long-term workforce planning.

Q13. On average, how long have employees been working at the company (based on hire date)?

```
SELECT ROUND(AVG(DATEDIFF(CURDATE(), hire_date))/ 365,2) as avg_tenure_years
FROM employees;
```

Business Questions & Uses

avg_tenure_years
6.62

Business Use: Helps management assess employee experience and retention patterns through workforce tenure.

Q14. How many new employees were hired each month (hiring trend)?

```
SELECT DATE_FORMAT(hire_date, '%Y-%m') as hire_month, COUNT(*) as new_hires
FROM employees
GROUP BY hire_month
ORDER BY hire_month;
```

hire_month	new_hires
2015-01	71
2015-02	77
2015-03	95
2015-04	78
2015-05	86
2015-06	80
2015-07	88
2015-08	80
2015-09	72
2015-10	75
2015-11	113
2015-12	84
2016-01	80

Business Use: Helps HR identify monthly hiring patterns to improve recruitment planning and resource allocation.

Q15. Which department offers the highest average salary?

```
SELECT d.department_name, ROUND(AVG(e.salary)) as avg_salary
FROM employees e
JOIN departments d
ON d.department_id = e.department_id
GROUP BY department_name
ORDER BY avg_salary DESC
LIMIT 1;
```

department_name	avg_salary
Operations	138354

Business Use: Helps management identify departments requiring higher salary budgets.

Q16. Which designation/job role earns the highest average salary?

```
SELECT designation, ROUND(AVG(salary)) as avg_salary
FROM employees
GROUP BY designation
ORDER BY avg_salary DESC
LIMIT 1;
```

Business Questions & Uses

designation	avg_salary
Procurement Executive	147337

Business Use: Helps HR identify job roles requiring higher compensation budgets and review pay structures.

Q17. Which employees are eligible for promotion based on high-performance ratings?

```
SELECT e.employee_id, e.employee_name, p.performance_rating
FROM employees e
JOIN performance p
ON e.employee_id = p.employee_id
WHERE p.performance_rating > 4.5
ORDER BY p.performance_rating DESC;
```

employee_id	employee_name	performance_rating
4438	Saanvi Singh	5.00
3985	Kavya Agarwal	5.00
7709	Sanjay Bose	5.00
4693	Kavya Joshi	5.00
1727	Vivaan Bose	5.00
3799	Priya Bose	5.00
6748	Rohan Agarwal	5.00
3949	Ananya Joshi	5.00
3401	Sanjay Chopra	5.00
5402	Swati Mehta	5.00
8293	Swati Gupta	5.00
498	Aditya Menon	5.00
9740	Arjun Bose	5.00
3148	Revansh Menon	5.00

Business Use: Helps HR identify potential promotion candidates using performance-based criteria.

Q18. How are performance ratings distributed across the company (how many employees fall in each rating range)?

```
SELECT CASE
        WHEN performance_rating <=1 THEN '0.0-1.0'
    WHEN performance_rating <=2 THEN '1.1-2.0'
    WHEN performance_rating <=3 THEN '2.1-3.0'
    WHEN performance_rating <=4 THEN '3.1-4.0'
    ELSE '4.1-5.0'
     END as perfomance_rating_range,
  COUNT(*) as num_of_employees
FROM performance
GROUP BY perfomance_rating_range
ORDER BY perfomance_rating_range;
```

perfomance_rating_range	num_of_employees
0.0-1.0	10
1.1-2.0	2544
2.1-3.0	2499
3.1-4.0	2440
4.1-5.0	2507

Business Questions & Uses

Business Use: Helps management identify whether the workforce has a healthy distribution of performance levels.

Q19. How has the employee headcount grown over time (year/month-wise)?

SELECT YEAR(hire_date) as hire_year , COUNT(*) as employees_headcount

FROM employees

GROUP BY hire_year

ORDER BY hire_year;

hire_year	employees_hired
2015	999
2016	971
2017	952
2018	995
2019	1028
2020	993
2021	1030
2022	1046
2023	1003
2024	983

Business Use: Helps HR analyze annual recruitment demand and identify changes in hiring activity over time.

Q20. How many employees are based in each city?

SELECT city, COUNT(*) as num_of_employees

FROM employees

GROUP BY city

ORDER BY num_of_employees DESC;

city	num_of_employees
Bhopal	583
Ahmedabad	580
Nagpur	578
Chandigarh	570
Kolkata	570
Jaipur	568
Lucknow	568
Indore	559
Delhi	552
Patna	552
Varanasi	551
Bengaluru	549
Chennai	549
Surat	547
Hyderabad	539
Mumbai	538
Kochi	536
Pune	511

Business Use: Helps management understand employee concentration by location for workforce and location planning.

Q21. Which department has the highest attrition rate?

SELECT d.department_name,

Business Questions & Uses

```
ROUND(SUM((CASE
        WHEN employment_status IN ('Resigned', 'Terminated')
  THEN 1 ELSE 0 END))/ COUNT(*) * 100,2) as attrition_rate
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
GROUP BY d.department_name
ORDER BY attrition_rate DESC;
```

department_name	attrition_rate
Information Technology	32.63
Quality Assurance	32.41
Procurement	32.19
Finance	30.47
Human Resources	30.19
Customer Support	29.62
Legal	29.31
Administration	28.81
Marketing	28.78
Operations	28.61
Research and Develop...	27.77
Sales	26.78

Business Use: Helps management identify departments with potential retention problems requiring further investigation.

Q22. What does the overall workforce summary look like (total headcount, active employees, attrition rate,

-- average salary, average rating — all in, one view)?

```
SELECT
        (SELECT COUNT(*) FROM employees) as total_headcount,
        (SELECT COUNT(*) FROM employees WHERE employment_status = 'Active') as active_employees,
        (SELECT ROUND(SUM(CASE WHEN employment_status IN ('Terminated', 'Resigned') THEN 1 ELSE 0
END)/COUNT(*) *100,2) FROM employees) as attrition_rate,
       (SELECT ROUND(AVG(salary)) FROM employees) as average_salary,
       (SELECT ROUND(AVG(performance_rating),2) FROM performance) as average_rating;
```

total_headcount	active_employees	attrition_rate	average_salary	average_rating
10000	7018	29.82	136302	2.99

Business Use: Provides management with a high-level workforce snapshot to support HR decision-making.

Key Takeaways

- Finance department has the largest workforce with 886 employees.
- Operations department has the highest average salary i.e. 1,38,354/-.
- Employee attrition is highest in IT department with 32.63%.
- 12.2% of employees fall within the highest performance category.

Conclusion

This project demonstrates the use of SQL to transform HR data into business-focused insights. The analysis can support workforce planning, compensation analysis, performance management, recruitment planning, and employee retention decisions.

SQL Skills Demonstrated

Area	Skills
Data Retrieval	SELECT, WHERE, filtering and sorting
Aggregation	COUNT, SUM, GROUP BY
Data Combination	JOIN
Conditional Logic	CASE WHEN
Date Analysis	Year/month extraction
Advanced SQL	CTEs and window functions